

| Database Systems |                                                                                                |                |
|------------------|------------------------------------------------------------------------------------------------|----------------|
| Lab #:           | 04                                                                                             |                |
| Lab Title:       | Relationships, Simple Query Designing and Advance Queries<br>(Custom Calculations in Queries). |                |
| Submitted by:    |                                                                                                |                |
| Names            |                                                                                                | Registration # |
| Sumaira Safeer   |                                                                                                | FA22-BCE-019   |

| Rubrics name & number   |                                                                                                                                                                                       |    |    |    | Marks  |          |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|--------|----------|
|                         |                                                                                                                                                                                       |    |    |    | In-Lab | Post-Lab |
| Engineering Knowledge   | <b>R2: Use of Engineering Knowledge and follow Experiment Procedures:</b><br>Ability to follow experimental procedures, control variables, and record procedural steps on lab report. |    |    |    |        |          |
| Problem Analysis        | <b>R6: Experimental Data Analysis:</b><br>Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.                               |    |    |    |        |          |
| Design                  | <b>R8: Best Coding Standards:</b><br>Ability to follow the coding standards and programming practices.                                                                                |    |    |    |        |          |
| Modern Tools Usage      | <b>R9: Understand Tools:</b> Ability to describe and explain the principles behind and applicability of engineering tools.                                                            |    |    |    |        |          |
| Individual and Teamwork | <b>R12: Individual Work Contributions:</b><br>Ability to carry out individual responsibilities.                                                                                       |    |    |    |        |          |
|                         | <b>R13: Management of Team Work:</b><br>Ability to appreciate, understand and work with multidisciplinary team members.                                                               |    |    |    |        |          |
| Rubrics #               | R2                                                                                                                                                                                    | R6 | R8 | R9 | R12    | R13      |
| In Lab                  |                                                                                                                                                                                       |    |    |    |        |          |
| Post- Lab               |                                                                                                                                                                                       |    |    |    |        |          |

## Lab No: 04

### Title: Relationships, Simple Query Designing and Advance Queries (Custom Calculations in Queries).

#### Objective:

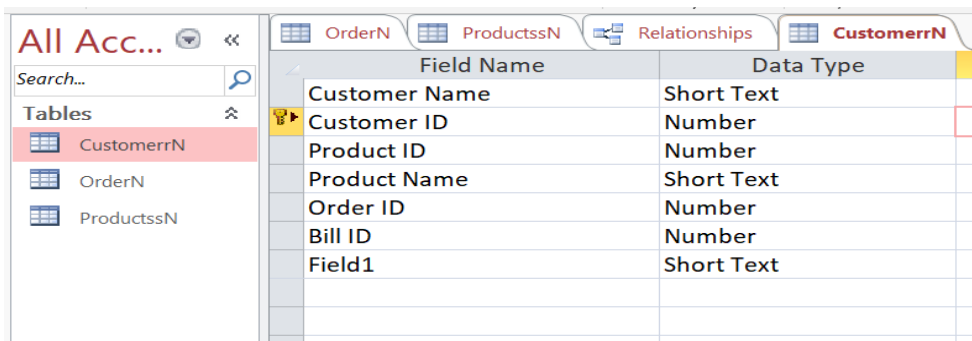
- How to create relationships between tables and how to design a query. Using a query makes it easier to view, add, delete, or change data in your database.  
The objective of query is:
- Find specific data quickly by filtering on specific criteria (conditions).
- Calculate or summarize data.
- Automate data management tasks, such as reviewing the most current data on a recurring basis.

#### In Lab Task:

##### Task 01:

Create a many to many relationships between Customer and Product as 1 Customer can places an order for more than one product and one product can be the result of more than one customers.

#### Design View:



| Field Name    | Data Type  |
|---------------|------------|
| Customer Name | Short Text |
| Customer ID   | Number     |
| Product ID    | Number     |
| Product Name  | Short Text |
| Order ID      | Number     |
| Bill ID       | Number     |
| Field1        | Short Text |

#### Datasheet View:

| All Acc... | OrderN        | ProductssN  | Relationships | CustomerN          |          |         |        |              |
|------------|---------------|-------------|---------------|--------------------|----------|---------|--------|--------------|
| Search...  | Customer Name | Customer ID | Product ID    | Product Name       | Order ID | Bill ID | Field1 | Click to Add |
| Tables     |               |             |               |                    |          |         |        |              |
| CustomerN  | Noreen zeer   | 347         | 206           | Electronics        | 1006     | 236     |        |              |
| OrderN     | Maria Milana  | 456         | 201           | Electronic Devices | 1001     | 234     |        |              |
| ProductssN | Farhan Faizan | 573         | 202           | Crafty products    | 1002     | 245     |        |              |
|            | Zoya Rohid    | 576         | 203           | MakeUp             | 1003     | 267     |        |              |
|            | Amina Khan    | 760         | 205           | Gitmub             | 1005     | 276     |        |              |
|            | Aleesha       | 785         | 204           | Atomic             | 1004     | 278     |        |              |
|            | *             | 0           | 0             |                    | 0        | 0       |        |              |

## Design View:

| All Acc... | OrderN       | ProductssN |  |  |
|------------|--------------|------------|--|--|
| Search...  | Field Name   | Data Type  |  |  |
| Tables     | Customer ID  | Number     |  |  |
| CustomerN  | Product Name | Short Text |  |  |
| OrderN     | Order ID     | Short Text |  |  |
| ProductssN |              |            |  |  |

## Datasheet View:

| All Acc... | OrderN      | ProductssN         |          |              |  |  |
|------------|-------------|--------------------|----------|--------------|--|--|
| Search...  | Customer ID | Product Name       | Order ID | Click to Add |  |  |
| Tables     |             |                    |          |              |  |  |
| CustomerN  | 456         | Electronic Devices | 1001     |              |  |  |
| OrderN     | 573         | Crafty products    | 1002     |              |  |  |
| ProductssN | 576         | MakeUp             | 1003     |              |  |  |
|            | 785         | Atomic             | 1004     |              |  |  |
|            | 760         | Get mub            | 1005     |              |  |  |
|            | 347         | Electronics        | 1006     |              |  |  |
|            | *           | 0                  |          |              |  |  |

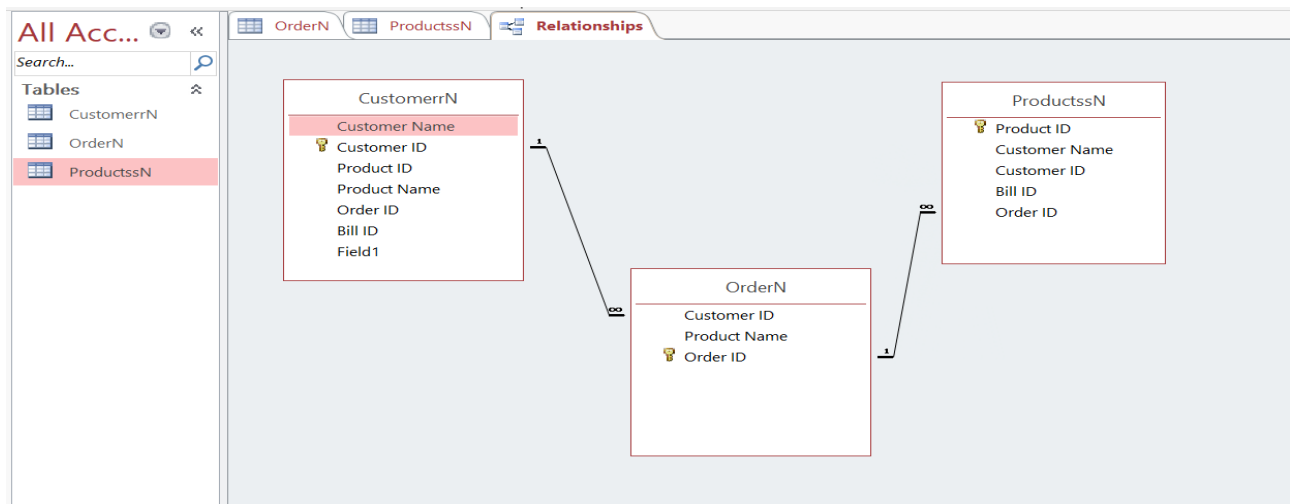
## Design View:

| All Acc... | OrderN        | ProductssN |  |  |
|------------|---------------|------------|--|--|
| Search...  | Field Name    | Data Type  |  |  |
| Tables     | Product ID    | Number     |  |  |
| CustomerN  | Customer Name | Short Text |  |  |
| OrderN     | Customer ID   | Number     |  |  |
| ProductssN | Bill ID       | Number     |  |  |
|            | Order ID      | Short Text |  |  |

## Datasheet View:

| Product ID | Customer Name | Customer ID | Bill ID | Order ID | Click to Add |
|------------|---------------|-------------|---------|----------|--------------|
| 201        | Maria Milana  | 456         | 234     | 1001     |              |
| 202        | Farhan Faizan | 573         | 245     | 1002     |              |
| 203        | Zoya Rohid    | 576         | 267     | 1003     |              |
| 204        | Aleesha       | 785         | 278     | 1004     |              |
| 205        | Amina Khan    | 760         | 276     | 1005     |              |
| 206        | Noreen zeer   | 347         | 236     | 1006     |              |
| 0          |               | 0           | 0       |          |              |

## Relationship:



## Task 02:

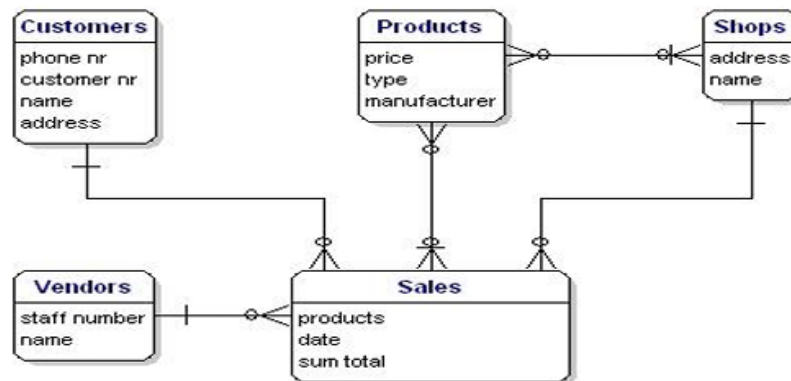
Design these relationships after creating the tables.

- Customers → Sales: 1 customer can buy something several times.
- Sales → Customers: 1 sale is always made by 1 customer at the time.
- Vendors → sales: 1 vendor can do multiple sales.
- Sales → vendor: 1 sale can be done by only 1 vendor at a time.
- Shops → Products: in 1 shop there are multiple products.
- Products → Shops: 1 product (type) can be sold in multiple shops.
- Shops → Sales: in 1 shop multiple sales can be made.
- Sales → Shops: 1 sale can only be made in 1 shop at the time.
- Products → Sales: 1 product (type) can be purchased in multiple sales.
- Sales → Products: 1 sale can exist out of multiple products.

### Relationships:

- Customers → Sales: 1: N (one to many)
- Sales → Products: M: N (many to many)
- Shops → Sales: 1: N (one to many)
- Shops → Products: M: N (many to many)
- Vendors → sales: 1: N (one to many)

**Final Relationship with keys and associative tables must be like this:**



**Design View:**

| All Acc...          |            | Shops Table | Vendors Table | Relationships | Custo |
|---------------------|------------|-------------|---------------|---------------|-------|
| Search...           |            |             |               |               |       |
| Tables              |            |             |               |               |       |
| Customers Table     |            |             |               |               |       |
| Products Table      |            |             |               |               |       |
| Sales Table         |            |             |               |               |       |
| SalesProducts Table |            |             |               |               |       |
| ShopProducts Table  |            |             |               |               |       |
| Shops Table         |            |             |               |               |       |
| Vendors Table       |            |             |               |               |       |
| Field Name          | Data Type  |             |               |               |       |
| Customer ID         | Number     |             |               |               |       |
| Customer Name       | Short Text |             |               |               |       |
| Contact Info        | Long Text  |             |               |               |       |

**Datasheet View:**

| All Acc...          |               | Shops Table           | Vendors Table | Relationships | Customers Table |
|---------------------|---------------|-----------------------|---------------|---------------|-----------------|
| Search...           |               |                       |               |               |                 |
| Tables              |               |                       |               |               |                 |
| Customers Table     |               |                       |               |               |                 |
| Products Table      |               |                       |               |               |                 |
| Sales Table         |               |                       |               |               |                 |
| SalesProducts Table |               |                       |               |               |                 |
| ShopProducts Table  |               |                       |               |               |                 |
| Shops Table         |               |                       |               |               |                 |
| Vendors Table       |               |                       |               |               |                 |
| Customer ID         | Customer Name | Contact Info          |               |               |                 |
| +                   | 10 Zuoshang   | Zuoshang@example.com  |               |               |                 |
| +                   | 20 John Doe   | Johndoe@example.com   |               |               |                 |
| +                   | 30 Jae Jo     | Jaejo@example.com     |               |               |                 |
| +                   | 40 Jane Smith | Janesmith@example.com |               |               |                 |
| +                   | 50 Tuhay Zoe  | Tuhayzoe@example.com  |               |               |                 |
| *                   | 0             |                       |               |               |                 |

## Design View:

| Field Name    | Data Type  |
|---------------|------------|
| Product ID    | Number     |
| Product Name  | Short Text |
| Product Price | Number     |

| General        |              |
|----------------|--------------|
| Field Size     | Long Integer |
| Format         |              |
| Decimal Places | Auto         |
| Input Mask     |              |
| Caption        |              |

## Datasheet View:

|   | Product ID | Product Name    | Product Price | Click to... |
|---|------------|-----------------|---------------|-------------|
| + | 1001       | Bike            | 2300000       |             |
| + | 1002       | machine         | 748900        |             |
| + | 1003       | Car             | 540000        |             |
| + | 1004       | Custume         | 5400          |             |
| + | 1005       | Automic Devices | 70000         |             |
| * | 0          |                 | 0             |             |

## Design View:

| Field Name  | Data Type |
|-------------|-----------|
| Sale ID     | Number    |
| Sale Date   | Date/Time |
| Customer ID | Number    |
| Vendor ID   | Number    |
| Shop ID     | Number    |
| product ID  | Number    |

## Datasheet View:

| All Acc...          | Shops Table | Vendors Table | Relationships | Customers Table | Products Table | Sales Table |
|---------------------|-------------|---------------|---------------|-----------------|----------------|-------------|
| Search...           | Sale ID     | Sale Date     | Customer ID   | Vendor ID       | Shop ID        | product ID  |
| Tables              | +           | 1             | 10/1/2024     | 10              | 11             | 1           |
| Customers Table     | +           | 2             | 10/2/2024     | 20              | 12             | 2           |
| Products Table      | +           | 3             | 10/3/2024     | 30              | 13             | 3           |
| Sales Table         | +           | 4             | 10/4/2024     | 40              | 14             | 3           |
| SalesProducts Table | +           | 5             | 10/5/2024     | 50              | 15             | 4           |
| ShopProducts Table  | *           | 0             |               | 0               | 0              | 0           |

## Design View:

| All Acc...          | Shops Table         | Vendors Table | Relationships | Customers |
|---------------------|---------------------|---------------|---------------|-----------|
| Search...           | Field Name          | Data Type     |               |           |
| Tables              | SalesProducts Table | Number        |               |           |
| Customers Table     | Sale ID             | Number        |               |           |
| Products Table      | Product ID          | Number        |               |           |
| Sales Table         |                     |               |               |           |
| SalesProducts Table |                     |               |               |           |
| ShopProducts Table  |                     |               |               |           |
| Shops Table         |                     |               |               |           |
| Vendors Table       |                     |               |               |           |

## Datasheet View:

| All Acc...          | Shops Table         | Vendors Table | Relationships | Customers Table | Products Table |
|---------------------|---------------------|---------------|---------------|-----------------|----------------|
| Search...           | SalesProducts Table | Sale ID       | Product ID    | Click to Add    |                |
| Tables              | 1000                | 1             | 1001          |                 |                |
| Customers Table     | 2000                | 2             | 1002          |                 |                |
| Products Table      | 3000                | 3             | 1003          |                 |                |
| Sales Table         | 4000                | 4             | 1004          |                 |                |
| SalesProducts Table | 5000                | 5             | 1005          |                 |                |
| ShopProducts Table  | *                   | 0             | 0             | 0               |                |
| Shops Table         |                     |               |               |                 |                |
| Vendors Table       |                     |               |               |                 |                |

## Design View:

| All Acc...          | Shops Table    | Vendors Table | Relationships | Custo |
|---------------------|----------------|---------------|---------------|-------|
| Search...           | Field Name     | Data Type     |               |       |
| Tables              | Shop ProductID | Number        |               |       |
| Customers Table     | Shop ID        | Number        |               |       |
| Products Table      | Product ID     | Number        |               |       |
| Sales Table         |                |               |               |       |
| SalesProducts Table |                |               |               |       |
| ShopProducts Table  |                |               |               |       |

## Datasheet View:

| All Acc...          |             |               |               |                 |
|---------------------|-------------|---------------|---------------|-----------------|
| Search...           | Shops Table | Vendors Table | Relationships | Customers Table |
| Tables              | Shop Produc | Shop ID       | Product ID    | Click to Add    |
| Customers Table     | 1000        | 1             | 1001          |                 |
| Products Table      | 2000        | 2             | 1002          |                 |
| Sales Table         | 3000        | 3             | 1003          |                 |
| SalesProducts Table | 4000        | 3             | 1004          |                 |
| ShopProducts Table  | 5000        | 4             | 1005          |                 |
| Shops Table         | *           | 0             | 0             | 0               |
| Vendors Table       |             |               |               |                 |

## Design View:

| All Acc...          |             |               |
|---------------------|-------------|---------------|
| Search...           | Shops Table | Vendors Table |
| Tables              | Field Name  | Data Type     |
| Customers Table     | Shop ID     | Number        |
| Products Table      | Shop Name   | Short Text    |
| Sales Table         | Location    | Short Text    |
| SalesProducts Table | Product ID  | Number        |
| ShopProducts Table  |             |               |
| Shops Table         |             |               |

## Datasheet View:

| All Acc...          |             |               |               |                 |
|---------------------|-------------|---------------|---------------|-----------------|
| Search...           | Shops Table | Vendors Table | Relationships | Customers Table |
| Tables              | Shop ID     | Shop Name     | Location      | Product ID      |
| Customers Table     | 1 shop 1    | mianwali      | 1001          | Click to Add    |
| Products Table      | 2 shop 2    | sanjwal       | 1002          |                 |
| Sales Table         | 3 shop 3    | bhakkar       | 1003          |                 |
| SalesProducts Table | 4 shop 4    | attock        | 1004          |                 |
| ShopProducts Table  | 5 shop 5    | sargodah      | 1005          |                 |
| Shops Table         | *           | 0             | 0             |                 |
| Vendors Table       |             |               |               |                 |

## Design View:



**All Acc...**

Search...

**Tables**

- bktblAuthor
- bktblPUBLISHER

**Queries**

- Query1

| Field Name      | Data Type  |
|-----------------|------------|
| ID              | AutoNumber |
| BOOK NAME       | Short Text |
| DATE OF PUBLISH | Short Text |

## Datasheet View:

**All Acc...**

Search...

**Tables**

- bktblAuthor
- bktblPUBLISHER

**Queries**

- Query1

| ID | BOOK NAME         | DATE OF PUBLISH | Click to Add |
|----|-------------------|-----------------|--------------|
| 1  | Vile Bodies       | 23-02-2003      |              |
| 2  | Wuthering heights | 14-02-2008      |              |
| 3  | A Little women    | 13-12-2017      |              |
| 4  | The greatest      | 29-05-2004      |              |
| 5  | A Scanner Darkly  | 14-02-2023      |              |
| 6  | Jane Eyre         | 10-09-2024      |              |
| *  | (New)             |                 |              |

## Design View:

**All Acc...**

Search...

**Tables**

- bktblAuthor
- bktblPUBLISHER

**Queries**

- Query1

| Field Name      | Data Type  |
|-----------------|------------|
| ID              | AutoNumber |
| BOOK NAME       | Short Text |
| DATE OF PUBLISH | Short Text |

| General    | Lookup       |
|------------|--------------|
| Field Size | Long Integer |
| New Values | Increment    |
| Format     |              |
| Caption    |              |
| Indexed    | No           |
| Text Align | General      |

## Datasheet View:

All Acc...

Search...

Tables

bktblAuthor

bktblPUBLISHER

Queries

Query1

Query1

bktblAuthor

bktblPUBLISHER

| ID     | BOOK NAME         | DATE OF PUBLISH | Click to Add |
|--------|-------------------|-----------------|--------------|
| 1      | Vile Bodies       | 23-02-2003      |              |
| 2      | Wuthering heights | 14-02-2008      |              |
| 3      | A Little women    | 13-12-2017      |              |
| 4      | The greatest      | 29-05-2004      |              |
| 5      | A Scanner Darkly  | 14-02-2023      |              |
| 6      | Jane Eyre         | 10-09-2024      |              |
| *(New) |                   |                 |              |

## Query:

| All Acc...     | Query1 | bktblAuthor | bktblPUBLISHER | Query2 |
|----------------|--------|-------------|----------------|--------|
| Search...      |        |             |                |        |
| Tables         |        |             |                |        |
| bktblAuthor    |        |             |                |        |
| bktblPUBLISHER |        |             |                |        |
| Queries        |        |             |                |        |
| Query1         |        |             |                |        |

| Field:    | ID                                  | Name                                | email                               | address                             | ID                                  | BOOK NAME                           | DATE OF PUBLISH                     |
|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Table:    | bktblAuthor                         | bktblAuthor                         | bktblAuthor                         | bktblAuthor                         | bktblPUBLISHER                      | bktblPUBLISHER                      | bktblPUBLISHER                      |
| Sort:     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |
| Show:     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Criteria: |                                     |                                     |                                     |                                     |                                     |                                     |                                     |
| or:       |                                     | "Bill Gates"                        |                                     |                                     |                                     |                                     |                                     |

## Query Result:

| ID | Name             | email              | BOOK NAME         | Contact No | Expr1005   | DATE OF PUBLISH |
|----|------------------|--------------------|-------------------|------------|------------|-----------------|
| 1  | G Thomsan        | gewqjk21@gmail.com | Vile Bodies       | 304281918  | 23-02-2003 | 23-02-2003      |
| 2  | RW Robot         | qwggu30@gmail.com  | Vile Bodies       | 301346932  | 23-02-2003 | 23-02-2003      |
| 3  | Robert frost     | gwqui49@gmail.com  | Vile Bodies       | 308233278  | 23-02-2003 | 23-02-2003      |
| 4  | Pugpne Goldstein | wekge58@gmail.com  | Vile Bodies       | 305231788  | 23-02-2003 | 23-02-2003      |
| 5  | Bil Gates        | ftyu67@gmail.com   | Vile Bodies       | 345677889  | 23-02-2003 | 23-02-2003      |
| 1  | G Thomsan        | gewqjk21@gmail.com | Wuthering heights | 304281918  | 14-02-2008 | 14-02-2008      |
| 2  | RW Robot         | qwggu30@gmail.com  | Wuthering heights | 301346932  | 14-02-2008 | 14-02-2008      |
| 3  | Robert frost     | gwqui49@gmail.com  | Wuthering heights | 308233278  | 14-02-2008 | 14-02-2008      |
| 4  | Pugpne Goldstein | wekge58@gmail.com  | Wuthering heights | 305231788  | 14-02-2008 | 14-02-2008      |
| 5  | Bil Gates        | ftyu67@gmail.com   | Wuthering heights | 345677889  | 14-02-2008 | 14-02-2008      |
| 1  | G Thomsan        | gewqjk21@gmail.com | A Little women    | 304281918  | 13-12-2017 | 13-12-2017      |
| 2  | RW Robot         | qwggu30@gmail.com  | A Little women    | 301346932  | 13-12-2017 | 13-12-2017      |
| 3  | Robert frost     | gwqui49@gmail.com  | A Little women    | 308233278  | 13-12-2017 | 13-12-2017      |
| 4  | Pugpne Goldstein | wekge58@gmail.com  | A Little women    | 305231788  | 13-12-2017 | 13-12-2017      |
| 5  | Bil Gates        | ftyu67@gmail.com   | A Little women    | 345677889  | 13-12-2017 | 13-12-2017      |
| 1  | G Thomsan        | gewqjk21@gmail.com | The greatest      | 304281918  | 29-05-2004 | 29-05-2004      |
| 2  | RW Robot         | qwggu30@gmail.com  | The greatest      | 301346932  | 29-05-2004 | 29-05-2004      |
| 3  | Robert frost     | gwqui49@gmail.com  | The greatest      | 308233278  | 29-05-2004 | 29-05-2004      |
| 4  | Pugpne Goldstein | wekge58@gmail.com  | The greatest      | 305231788  | 29-05-2004 | 29-05-2004      |
| 5  | Bil Gates        | ftyu67@gmail.com   | The greatest      | 345677889  | 29-05-2004 | 29-05-2004      |
| 1  | G Thomsan        | gewqjk21@gmail.com | A Scanner Darkly  | 304281918  | 14-02-2023 | 14-02-2023      |
| 2  | RW Robot         | qwggu30@gmail.com  | A Scanner Darkly  | 301346932  | 14-02-2023 | 14-02-2023      |
| 3  | Robert frost     | gwqui49@gmail.com  | A Scanner Darkly  | 308233278  | 14-02-2023 | 14-02-2023      |

## Task 04:

Create a new query that displays title ID, title name, the publisher's name, and the author's names.

## Design View:

| Field Name | Data Type  |
|------------|------------|
| ID         | Number     |
| Title name | Short Text |
| Reference  | Short Text |

## Datasheet View:

All Acc... << >>

Search...

Tables

- Author and Pub
- Title

Queries

- Query1

| ID  | Title name            | Reference         | Click to Add |
|-----|-----------------------|-------------------|--------------|
| 101 | To Kill a Mockingbird | Harper Lee        |              |
| 102 | Pride and Prejudice   | Jane Austen       |              |
| 103 | Vile Bodies           | J.D. Salinger     |              |
| 104 | Titlle homen          | Paulo Coelho      |              |
| 105 | The Hobbit            | J.R.R. Tolkien    |              |
| 106 | The Alchemist         | Jane Eyre         |              |
| 107 | The Great Gatsby      | F. Scott Fitzger: |              |
| *   | 0                     |                   |              |

## Design View:

All Acc... << >>

Search...

Tables

- Author and Pub
- Title

Queries

- Query1

| Field Name     | Data Type  |
|----------------|------------|
| ID             | Number     |
| Author Name    | Short Text |
| Publisher Name | Short Text |

## Datasheet View:

All Acc... << >>

Search...

Tables

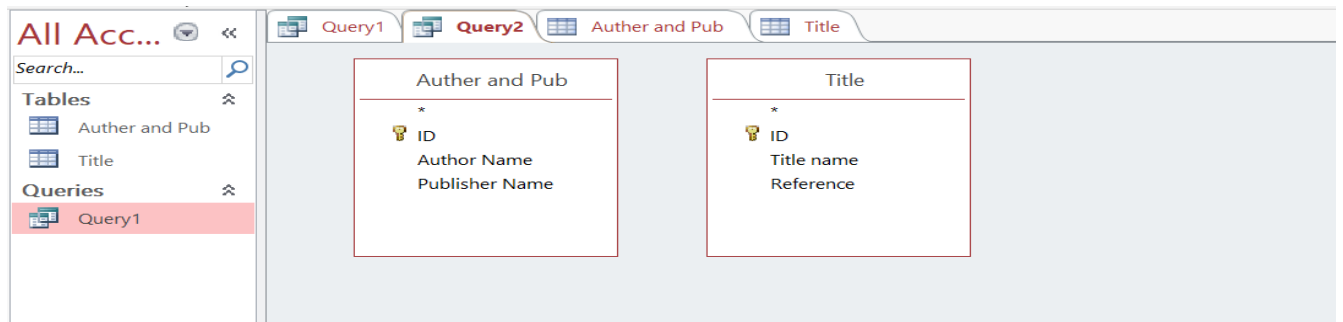
- Author and Pub
- Title

Queries

- Query1

| ID  | Author Name         | Publisher Name          | Click to Add |
|-----|---------------------|-------------------------|--------------|
| 110 | George Orwell       | Secker & Warburg        |              |
| 120 | Harper Lee          | J.B. Lippincott & Co.   |              |
| 130 | F. Scott Fitzgerald | Charles Scribner's Sons |              |
| 140 | Jane Austen         | T. Egerton, Whitehall   |              |
| 150 | Herman Melville     | Harper & Brothers       |              |
| 160 | J.K. Rowling        | Bloomsbury Publisher    |              |
| 170 | Paulo Coelho        | Harper & Row            |              |
| *   | 0                   |                         |              |

## Query:



|           |                                     |                                     |                                     |                                     |                                     |                                     |
|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Field:    | ID                                  | Author Name                         | Publisher Name                      | ID                                  | Title name                          | Reference                           |
| Table:    | Author and Pub                      | Author and Pub                      | Author and Pub                      | Title                               | Title                               | Title                               |
| Sort:     |                                     |                                     |                                     |                                     |                                     |                                     |
| Show:     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Criteria: |                                     |                                     |                                     |                                     |                                     |                                     |
| or:       |                                     | "George Orwell "                    | " Secker & Warburg"                 |                                     | "Pride and Prejudice"               |                                     |

## Query Result:

| ID  | Title name            | Author Name         | Publisher Name       |  |  |
|-----|-----------------------|---------------------|----------------------|--|--|
| 105 | The Hobbit            | George Orwell       | Secker & Warburg     |  |  |
| 106 | The Alchemist         | George Orwell       | Secker & Warburg     |  |  |
| 107 | The Great Gatsby      | George Orwell       | Secker & Warburg     |  |  |
| 101 | To Kill a Mockingbird | George Orwell       | Secker & Warburg     |  |  |
| 102 | Pride and Prejudice   | George Orwell       | Secker & Warburg     |  |  |
| 103 | Vile Bodies           | George Orwell       | Secker & Warburg     |  |  |
| 104 | Titlle homen          | George Orwell       | Secker & Warburg     |  |  |
| 105 | The Hobbit            | Harper Lee          | J.B. Lippincott & Co |  |  |
| 106 | The Alchemist         | Harper Lee          | J.B. Lippincott & Co |  |  |
| 107 | The Great Gatsby      | Harper Lee          | J.B. Lippincott & Co |  |  |
| 101 | To Kill a Mockingbird | Harper Lee          | J.B. Lippincott & Co |  |  |
| 102 | Pride and Prejudice   | Harper Lee          | J.B. Lippincott & Co |  |  |
| 103 | Vile Bodies           | Harper Lee          | J.B. Lippincott & Co |  |  |
| 104 | Titlle homen          | Harper Lee          | J.B. Lippincott & Co |  |  |
| 105 | The Hobbit            | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 106 | The Alchemist         | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 107 | The Great Gatsby      | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 101 | To Kill a Mockingbird | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 102 | Pride and Prejudice   | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 103 | Vile Bodies           | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 104 | Titlle homen          | F. Scott Fitzgerald | Charles Scribner's S |  |  |
| 105 | The Hobbit            | Jane Austen         | T. Egerton, Whiteha  |  |  |
| 106 | The Alchemist         | Jane Austen         | T. Egerton, Whiteha  |  |  |

## Task 05:

The expression we created for profit is too simplified. Modify it so that the profit is calculated as:  $((\text{sales} * \text{price}) - \text{advance}) * (1 - \text{royalty rate})$ .

## Design View:

| All Acc... | profit          | query1       |
|------------|-----------------|--------------|
| Search...  | Field Name      | Data Type    |
| Tables     | ID              | AutoNumber   |
| profit     | Sales           | Currency     |
| Queries    | Price           | Currency     |
| query1     | Advance Payment | Currency     |
|            | Royalty Rate    | Large Number |

## Datasheet View:

| All Acc... | profit | ID     | Sales      | Price      | Advance Payment | Royalty Rate | Click to Add |
|------------|--------|--------|------------|------------|-----------------|--------------|--------------|
| Search...  |        | 5      | \$35.00    | \$12.06    | \$2.99          | 56670        |              |
| Tables     |        | 8      | \$230.12   | \$167.00   | \$23.04         | 35676        |              |
| profit     |        | 7      | \$231.12   | \$27.03    | \$18.00         | 24450        |              |
| Queries    |        | 3      | \$340.00   | \$78.68    | \$9.35          | 25667        |              |
| query1     |        | 2      | \$345.00   | \$243.00   | \$46.00         | 44470        |              |
|            |        | 4      | \$500.50   | \$367.00   | \$13.01         | 55667        |              |
|            |        | 6      | \$678.09   | \$457.00   | \$31.98         | 55543        |              |
|            |        | 1      | \$3,000.00 | \$8,388.00 | \$64.00         | 56350        |              |
|            |        | *(New) | \$0.00     | \$0.00     | \$0.00          | 0            |              |

## Entering Parameter Value:

| All Acc... | profit | ID     | Sales      | Price      | Advance Payment | Royalty Rate | Click to Add |
|------------|--------|--------|------------|------------|-----------------|--------------|--------------|
| Search...  |        | 5      | \$35.00    | \$12.06    |                 |              |              |
| Tables     |        | 8      | \$230.12   | \$167.00   |                 |              |              |
| profit     |        | 7      | \$231.12   | \$27.03    |                 |              |              |
| Queries    |        | 3      | \$340.00   | \$78.68    |                 |              |              |
| query1     |        | 2      | \$345.00   | \$243.00   |                 |              |              |
|            |        | 4      | \$500.50   | \$367.00   | \$13.01         | 55667        |              |
|            |        | 6      | \$678.09   | \$457.00   | \$31.98         | 55543        |              |
|            |        | 1      | \$3,000.00 | \$8,388.00 | \$64.00         | 56350        |              |
|            |        | *(New) | \$0.00     | \$0.00     | \$0.00          | 0            |              |

Enter Parameter Value

Sales Price

37.45

OK

Cancel

## Calculated Profit :

| All Acc... | profit | query1 | sales      | price      | Advance Pay | Royalty Rate | Expr1      |
|------------|--------|--------|------------|------------|-------------|--------------|------------|
| Search...  |        |        | \$35.00    | \$12.06    | \$2.99      | 56670        | -6345540   |
| Tables     |        |        | \$230.12   | \$167.00   | \$23.04     | 35676        | -30782102  |
| profit     |        |        | \$231.12   | \$27.03    | \$18.00     | 24450        | -16481071  |
| Queries    |        |        | \$340.00   | \$78.68    | \$9.35      | 25667        | -8987142   |
| query1     |        |        | \$345.00   | \$243.00   | \$46.00     | 44470        | -76606746  |
|            |        |        | \$500.50   | \$367.00   | \$13.01     | 55667        | -27112249  |
|            |        |        | \$678.09   | \$457.00   | \$31.98     | 55543        | -66519932  |
|            |        |        | \$3,000.00 | \$8,388.00 | \$64.00     | 56350        | -135057283 |
|            |        |        | \$0.00     | \$0.00     | \$0.00      | 0            |            |

## Task 06:

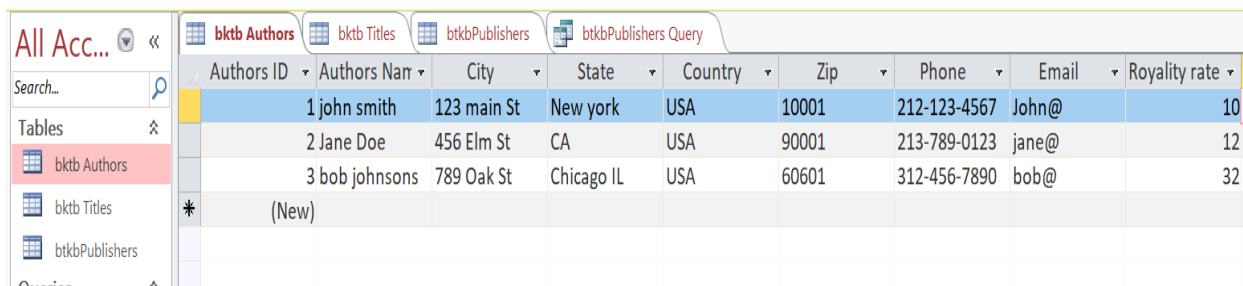
Combine both fields (price and sales) in one column with a space.

## Design View:



| Field Name   | Data Type  |
|--------------|------------|
| Authors ID   | AutoNumber |
| Authors Name | Short Text |
| City         | Short Text |
| State        | Short Text |
| Country      | Short Text |
| Zip          | Short Text |
| Phone        | Short Text |
| Email        | Short Text |
| Royalty rate | Number     |

## Datasheet View:



| Authors ID | Authors Name | City        | State      | Country | Zip   | Phone        | Email | Royalty rate |
|------------|--------------|-------------|------------|---------|-------|--------------|-------|--------------|
| 1          | john smith   | 123 main St | New york   | USA     | 10001 | 212-123-4567 | John@ | 10           |
| 2          | Jane Doe     | 456 Elm St  | CA         | USA     | 90001 | 213-789-0123 | jane@ | 12           |
| 3          | bob johnsons | 789 Oak St  | Chicago IL | USA     | 60601 | 312-456-7890 | bob@  | 32           |
| *          | (New)        |             |            |         |       |              |       |              |

## Design View:



| Field Name  | Data Type  |
|-------------|------------|
| TitleID     | AutoNumber |
| PublisherID | Number     |
| TitleName   | Short Text |
| Type        | Short Text |
| Pages       | Number     |
| Price       | Currency   |
| Sales       | Number     |
| PubDate     | Date/Time  |
| Contract    | Short Text |

## Datasheet View:

| All Acc...                                                   |                 |          |       |         |       |          |          |         |          |  |
|--------------------------------------------------------------|-----------------|----------|-------|---------|-------|----------|----------|---------|----------|--|
| bktb Authors bktb Titles btkbPublishers btkbPublishers Query |                 |          |       |         |       |          |          |         |          |  |
| PublisherID                                                  | TitleName       | Type     | Pages | Price   | Sales | PubDate  | Contract | Advance | AuthorID |  |
| 1                                                            | The Great Nove  | Fiction  | 300   | \$25.00 | 5000  | 1/1/2020 | yes      | 10000   | 1        |  |
| 2                                                            | Mystory Book    | Thriller | 250   | \$20.00 | 3000  | 1/6/2019 | No       | 0       | 2        |  |
| 3                                                            | Science fiction | Sci-Fic  | 400   | \$30.00 | 7000  | 1/9/2020 | yes      | 15000   | 1        |  |
| *                                                            |                 |          |       |         |       |          |          |         |          |  |

## Design View:

All Acc...

Search...

Tables

bktb Authors

bktb Titles

btkbPublishers

Queries

btkbPublishers Qu...

bktb Authorsbktb TitlesbtkbPublishersbtkbPublishers Query

| Field Name    | Data Type  |
|---------------|------------|
| PublisherID   | AutoNumber |
| PublisherName | Short Text |
| Address       | Short Text |
| City          | Short Text |
| State         | Short Text |
| Country       | Short Text |
| Zip           | Short Text |
| Phone         | Long Text  |
| Email         | Long Text  |

General

Lookup

|            |              |
|------------|--------------|
| Field Size | Long Integer |
| New Values | Increment    |
| Format     |              |

## Datasheet View:

| bktb Authors bktb Titles btkbPublishers btkbPublishers Query |                |                |          |       |         |       |              |               |              |  |
|--------------------------------------------------------------|----------------|----------------|----------|-------|---------|-------|--------------|---------------|--------------|--|
| PublisherID                                                  | PublisherName  | Address        | City     | State | Country | Zip   | Phone        | Email         | Click to Add |  |
| 1                                                            | Penguin Rando  | 375 Hudson St  | New york | NY    | USA     | 10014 | 212-366-2000 | info@penguin  |              |  |
| 2                                                            | Hachette Book  | 1290 Avenue of | New york | NY    | USA     | 10104 | 212-698-7000 | info@hachette |              |  |
| 3                                                            | Harper Collins | 195 Broadway   | New york | NY    | USA     | 10007 | 212-207-7000 | info@harper   |              |  |
| 4                                                            |                |                |          |       |         |       |              |               |              |  |
| *                                                            | (New)          |                |          |       |         |       |              |               |              |  |

**All Acc...** Search...

**Tables**

- bktb Authors
- bktb Titles
- btkbPublishers

**Queries**

- btkbPublishers Qu...

**bktb Authors**

- Authors ID
- Authors Name
- City
- State
- Country
- Zip
- Phone
- Email
- Royalty rate

**bktb Titles**

- TitleID
- PublisherID
- TitleName
- Type
- Pages
- Price
- Sales
- PubDate
- Contract

**btkbPublishers**

- PublisherID
- PublisherName
- Address
- City
- State
- Country
- Zip
- Phone
- Email

**Query1**

| Field:    | TitleName                           | Price                               | Sales                               | Expr1: [price]*[Sales]              |
|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Table:    | bktb Titles                         | bktb Titles                         | bktb Titles                         |                                     |
| Sort:     |                                     |                                     |                                     |                                     |
| Show:     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Criteria: |                                     |                                     |                                     |                                     |
| or:       |                                     |                                     |                                     |                                     |

## Final Result:

**All Acc...** Search...

**Tables**

- bktb Authors
- bktb Titles
- btkbPublishers

**Queries**

- btkbPublishers Qu...
- Query1

**Query1**

| TitleName       | Price   | Sales | Expr1        |
|-----------------|---------|-------|--------------|
| The Great Nove  | \$25.00 | 5000  | \$125,000.00 |
| The Great Nove  | \$25.00 | 5000  | \$125,000.00 |
| The Great Nove  | \$25.00 | 5000  | \$125,000.00 |
| Mystery Book    | \$20.00 | 3000  | \$60,000.00  |
| Mystery Book    | \$20.00 | 3000  | \$60,000.00  |
| Mystery Book    | \$20.00 | 3000  | \$60,000.00  |
| Science fiction | \$30.00 | 7000  | \$210,000.00 |
| Science fiction | \$30.00 | 7000  | \$210,000.00 |
| Science fiction | \$30.00 | 7000  | \$210,000.00 |

## Post lab Tasks

### Postlab Task 1:

Only show “title\_name” and “type” records with royalty rates less than 0.08 or with advances that are less than 30000. Do not show the advance and royalty rates fields in the query.

### Query Check:

The screenshot shows the Microsoft Access interface with a table named 'bktbltitle' and a query named 'Query1'. The table has the following data:

| title_ID | Title_name | Type       | Pages | Price   | Sales | Advance | Royalty rate | Publish_ID |
|----------|------------|------------|-------|---------|-------|---------|--------------|------------|
| 1        | how about  | Psychology | 107   | \$13.00 | 100   | 60000   | 0.09         | 12         |
| 2        | Data base  | Computer   | 510   | \$4.00  | 700   | 20000   | 0.08         | 13         |
| 3        | CA         | Psychology | 77    | \$20.00 | 550   | 45000   | 1.05         | 14         |
| 6        | computer   | Biology    | 23    | \$27.00 | 400   | 14000   | 0.02         | 1          |
| 7        | IT based   | Computer   | 10    | \$8.00  | 800   | 14567   | 0.09         | 3          |
| 8        | About PMO  | Biology    | 9     | \$29.00 | 200   | 22000   | 0.06         | 6          |

The query 'Query1' is shown below the table, with a criteria filter on the 'Royalty rate' field set to '< 0.08'.

The screenshot shows the results of the query 'Query1'. The query returns the following data:

| title_ID | Title_name |
|----------|------------|
| 2        | Data base  |
| 6        | computer   |
| 7        | IT based   |
| 8        | About PMO  |

### PostlabTask 2: IFF Statement

We would like our new field to contain the value “has a contract” if the value of the sales field in bktblTitles is greater than 500, and “no contract” if that value is less than and equal to 500. Write the expression in expression builder.

### Design View:

|           |                                     |                                     |                                     |                                                         |
|-----------|-------------------------------------|-------------------------------------|-------------------------------------|---------------------------------------------------------|
| Field:    | title_ID                            | Title_name                          | Sales                               | Filed: If([Sales]>"500","has a contract","no contract") |
| Table:    | bktbltitle                          | bktbltitle                          | bktbltitle                          |                                                         |
| Sort:     |                                     |                                     |                                     |                                                         |
| Show:     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>                     |
| Criteria: |                                     |                                     |                                     |                                                         |
| or:       |                                     |                                     |                                     |                                                         |

### Expression builder Check:

| Query1 |          | bktbltitle |       | Query2         |  |
|--------|----------|------------|-------|----------------|--|
|        | title_ID | Title_name | Sales | Filed          |  |
|        | 1        | how about  | 100   | no contract    |  |
|        | 2        | Data base  | 700   | has a contract |  |
|        | 3        | CA         | 550   | has a contract |  |
|        | 6        | computer   | 400   | no contract    |  |
|        | 7        | IT based   | 800   | has a contract |  |
|        | 8        | About PMO  | 200   | no contract    |  |
| *      |          |            |       |                |  |

### Postlab Task 3:

Modify the expression name so that it is called "Contract" in Task 2.

|           |                                     |                                     |                                     |                                                            |
|-----------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------------------------------|
| Field:    | title_ID                            | Title_name                          | Sales                               | Contract: If([Sales]>"500","has a contract","no contract") |
| Table:    | bktbltitle                          | bktbltitle                          | bktbltitle                          |                                                            |
| Sort:     |                                     |                                     |                                     |                                                            |
| Show:     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>                        |
| Criteria: |                                     |                                     |                                     |                                                            |
| or:       |                                     |                                     |                                     |                                                            |

|  Query1 |  bktbltitle |  Query2 |                |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------|
| title_ID                                                                                   | Title_name                                                                                     | Sales                                                                                      | Contract       |
| 1                                                                                          | how about                                                                                      | 100                                                                                        | no contract    |
| 2                                                                                          | Data base                                                                                      | 700                                                                                        | has a contract |
| 3                                                                                          | CA                                                                                             | 550                                                                                        | has a contract |
| 6                                                                                          | computer                                                                                       | 400                                                                                        | no contract    |
| 7                                                                                          | IT based                                                                                       | 800                                                                                        | has a contract |
| 8                                                                                          | About PMO                                                                                      | 200                                                                                        | no contract    |
| *                                                                                          |                                                                                                |                                                                                            |                |

### Conclusion:

At the end of the lab, we are able to:

- Know about the query designing in MS Access.
- Implementing queries in Database.
- Using “Query Wizard” panel of MS Access.

